Year 2 · Chapter 1 · Algebraic Methods

Starter

1) Factorise each polynomial:
   a) $x^2 - 6x + 5$
   b) $x^2 - 16$
   c) $9x^2 - 25$
2) Simplify fully the following algebraic fractions:
   a) $\frac{x^2-9}{x^2+9x+18}$
   b) $\frac{2x^2+5x-12}{6x^2-7x-3}$
   c) $\frac{x^2-x-30}{-x^2+3x+18}$
3) For any integers $n$and m, decide whether the following will always be odd, always be even, or could be either.
   a) $8n$
   b) $n - m$
   c) $3m$
   d) $2n - 5$

Chapter Overview

Apart from the initial portion on proof, this chapter concerns manipulation of algebraic fractions.

1: Proof By Contradiction

Prove by contradiction that $\sqrt{5}$ is an irrational number.

2: $+ \div \times -$ Algebraic Fractions

Express

$$\frac{3x+5}{x^2+x-12} - \frac{2}{x-3}$$

as a single fraction in its simplest form.

3: Express a fraction using partial fractions.

Express $\frac{2+3x}{(x+3)(x-1)}$ in the form $\frac{A}{x+3} + \frac{B}{x-1}$ where A and B are constants.

4: Divide algebraic expressions, and convert an improper fraction into partial fraction form.

Given that

$$\frac{x^3+x^2+7}{x-3} \equiv Ax^2 + Bx + C + \frac{D}{x-3}$$

find the values of A, B, C and D.

1: Proof By Contradiction

✎ To prove a statement is true by contradiction:
- **Assume** that the statement is in fact **false**.
- Prove that this would **lead to a contradiction**.
- Therefore we were wrong in assuming the statement was false, and therefore it must be true.

Prove that there is no greatest odd integer.

? Assumption

? Show contradiction

? Conclusion

How to structure/word proof:
1. "Assume that *[negation of statement]*."
2. [*Reasoning followed by…*] "*This contradicts the assumption that…*" or "*This is a contradiction*".
3. "Therefore [*restate original statement*]."

Negating the original statement 5

The first part of a proof by contradiction requires you to negate the original statement.
What is the negation of each of these statements? *(Click to choose)*

"There are infinitely many prime numbers."	"All Popes are Catholic."	"If it is raining, my garden is wet."
"There are infinitely many non-prime (i.e. composite) numbers."	"There exists a Pope who is not Catholic."	"It is not raining and my garden is dry."
"There are finitely many prime numbers."	"No Popes are Catholic."	"It is not raining and my garden is wet."
"There are finitely many non-composite numbers."	"Your Teacher is the Pope."	"It is raining and my garden is not wet."

Comments: The negation of "all are" is not "none are". So the negation of "everyone likes green" wouldn't be "no one likes green", but: "not everyone likes green". Do not confuse a 'negation' with the 'opposite'.

Comments: If you have a conditional statement like "*If A then B*", then the negation is "*A and not B*", i.e. the condition is true, but the conclusion is false/negated.

More Examples 6

Prove by contradiction that if n^2 is even, then n must be even.

? Assumption

? Show contradiction

? Conclusion

Remember the negation of "if A then B" is "A and not B".

More Examples

Prove by contradiction that $\sqrt{2}$ is an irrational number.

1

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A **rational number** is one that can be expressed in the form $\frac{a}{b}$ where a, b are integers.
An **irrational** number cannot be expressed in this form, e.g. $\pi, e, \sqrt{3}$.
The set of all rational numbers is $\mathbb{Q}$ (real numbers: $\mathbb{R}$, natural numbers: $\mathbb{N}$, integers: $\mathbb{Z}$).

This is the standard proof for the irrationality of $\sqrt{2}$.
An Alternative Proof is below:

$2b^2 = a^2$
If a number is square then the powers in the prime factorisation are even. The power of 2 on the RHS is therefore even, but odd on the LHS (due to the extra 2). This is a contradiction.

More Examples

Prove by contradiction that there are infinitely many prime numbers.

? Assumption

? Show contradiction

? Conclusion

This proof is courtesy of Euclid, and is one of the earliest known proofs.

5 a Prove that if ab is an irrational number then at least one of a and b is an irrational number.

(3 marks)

1

b Prove that if $a + b$ is an irrational number then at least one of a and b is an irrational number.

(

2

6 Use proof by contradiction to show that there exist no integers a and b for which $21a + 14b = 1$.

Answer

7 a Prove by contradiction that if n^2 is a multiple of 3, n is a multiple of 3. **(3 marks)**

Answer

b Hence prove by contradiction that $\sqrt{3}$ is an irrational number. **(3 marks)**

Answer

8 Use proof by contradiction to prove the statement:
'There are no integer solutions to the equation
$x^2 - y^2 = 2$'

Answer

Prove by contradiction that $\sqrt[3]{2}$ is irrational.

Answer

Extension

[STEP I 2008 Q1] What does it mean to say that a number x is rational?

Prove by contradiction statements A and B below, where p and q are real numbers.

A: If pq is irrational, then at least one of p and q is irrational.

B: If $p+q$ is irrational, then at least one of p and q is irrational.

Disprove by means of a counterexample statement C below, where p and q are real numbers.

C: If p and q are irrational, then $p+q$ is irrational.

If the numbers e, π, π^2, e^2 and $e\pi$ are irrational, prove that at most one of the numbers $\pi+e, \pi-e, \pi^2-e^2, \pi^2+e^2$ is rational.

Solutions on next slide.

Note: e is Euler's Number and you will encounter it later at A Level. But for the moment, you only need to know it is an irrational number, like π.

What does it mean to say that a number x is irrational*?*

1

Prove by contradiction statements A and B below, where p and q are real numbers.

***A:** If pq is irrational, then at least one of p and q is irrational.*

***B:** If $p+q$ is irrational, then at least one of p and q is irrational.*

2

Disprove by means of a counterexample statement C below, where p and q are real numbers.

***C:** If p and q are irrational, then $p+q$ is irrational.*

3

If the numbers e, π, π^2, e^2 *and* $e\pi$ *are irrational, prove that at most one of the numbers* $\pi + e$, $\pi - e$, $\pi^2 - e^2$, $\pi^2 + e^2$ *is rational.*

Answer

2a: Multiplying/Dividing Algebraic Fractions 18

As your saw at GCSE level, multiplying algebraic fractions is no different to multiply numeric fractions. You may however need to **cancel common factors, by factorising where possible**.

$\frac{a}{b} \times \frac{c}{a} = \mathbf{\frac{ac}{ab}}$ 1 $= \mathbf{\frac{3(x+1)}{2(x+1)(x-1)}} = \mathbf{\frac{3}{2(x-1)}}$

2

To divide by a fraction, **multiply by the reciprocal of the second fraction**.

$\frac{p}{q} \div \frac{r}{q} = \mathbf{\frac{p}{q}} \times$ 3 $\mathbf{-4)} = \mathbf{\frac{x-4}{3}}$

$\frac{x+2}{x+4} \div \frac{3x+6}{x^2-16}$

4

Test Your Understanding

19

$\frac{x+3}{5} \times \frac{10}{x^2-9}$

$\frac{x^2+x}{y} \div \frac{x^2-x-2}{y^2}$

1

2

Common student "Crime against Mathematics":

$\frac{x^2+y}{2y} \not\equiv \frac{x^2}{2}$

3

Exercise 1B (page 6-7)

20

3 Show that $\frac{x^2-64}{x^2-36} \div \frac{64-x^2}{x^2-36} = -1$

4 Show that $\frac{2x^2-11x-40}{x^2-4x-32}\times\frac{x^2+8x+16}{6x^2-3x-45}\div\frac{8x^2+20x-48}{10x^2-45x+45}=\frac{a}{b}$ and find the values of the constants a and b, where a and b are integers.

To add/subtract two fractions, find a common denominator.
This can be achieved by multiplying the denominators and cross-multiplying the numerators:

$$\frac{3}{x+1}-\frac{2}{x+2}=\frac{\mathbf{3(x+2)-2(x+1)}}{\mathbf{(x+1)(x+2)}}=\frac{\mathbf{x+4}}{\mathbf{(x+1)(x+2)}}$$

However, often we should see if first if there are common factors in the denominator **to avoid multiplying unnecessarily**:

$$\frac{3}{x+1}-\frac{4x}{x^2-1}$$
$$=\frac{3}{x+1}-\frac{4x}{(x+1)(x-1)}$$
$$=\frac{3(x-1)}{(x+1)(x-1)}-\frac{4x}{(x+1)(x-1)}$$
$$=\frac{-x-3}{(x+1)(x-1)}$$

1

2

[Edexcel C3 June 2013(R) Q1]

Express

$$\frac{3x+5}{x^2+x-12} - \frac{2}{x-3}$$

as a single fraction in its simplest form.

1

Express the following as a single fraction, giving your answer in its simplest form.

$$\frac{10x+4}{3x^2+4x+1} - \frac{3}{x+1}$$

2

Complete Exercise 1C, textbook page 8.

Open Resources → Book chapter to view the questions.

Partial Fractions

If the **denominator is a product of a linear terms**, it can be split into the sum of 'partial fractions', where **each denominator is a single linear term**.

$$\frac{6x-2}{(x-3)(x+1)} \equiv \frac{A}{x-3} + \frac{B}{x+1}$$

Notation reminder: $\equiv$ means 'equivalent/identical to', and indicates that both sides are equal for **all** values of x.

Method 1: Substitution

Method 1

Method 2: Comparing Coefficients

Method 2

Further Example

Given that $\frac{6x^2+5x-2}{x(x-1)(2x+1)} \equiv \frac{A}{x} + \frac{B}{x-1} + \frac{C}{2x+1}$, find the values of the constants A, B, C.

Answer

Express $\dfrac{5x+3}{(2x-3)(x+2)}$ in partial fractions.

(3)

Answer

4 Express $\dfrac{6x+1}{x^2+2x-15} - \dfrac{4}{x-3}$ as a single fraction in its simplest form.

Answer

6 Express $\dfrac{4(2x-1)}{36x^2-1} + \dfrac{7}{6x-1}$ as a single fraction in its simplest form.

Answer

7 $g(x) = x + \dfrac{6}{x+2} + \dfrac{36}{x^2-2x-8}, x \in \mathbb{R}, x \neq -2, x \neq 4$

a Show that $g(x) = \dfrac{x^3-2x^2-2x+12}{(x+2)(x-4)}$

b Using algebraic long division, or otherwise, further show that $g(x) = \dfrac{x^2-4x+6}{x-4}$

a 1

b 2

Repeated linear factors 31

Suppose we wished to express $\frac{2x+1}{(x+1)^2}$ as $\frac{A}{x+1}+\frac{B}{x+1}$. What's the problem?

Explanation

Split $\frac{11x^2+14x+5}{(x+1)^2(2x+1)}$ into partial fractions.

$$\frac{11x^2+14x+5}{(x+1)^2(2x+1)} \equiv \frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{2x+1}$$

The problem is resolved by having the factor **both squared and non-squared** (explanation of why we do this at the end of these slides).

Working

Test Your Understanding 32

$$\frac{9x^2}{(x-1)^2(2x+1)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(2x+1)}.$$

Find the values of the constants A, B and C. **(4)**

Answer

1 $f(x) = \dfrac{3x^2 + x + 1}{x^2(x + 1)}$, $x \neq 0$, $x \neq -1$

Given that $f(x)$ can be expressed in the form $\dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x + 1}$, find the values of A, B and C.

Answer

4 Show that $\dfrac{5x^2 - 2x - 1}{x^3 - x^2}$ can be written in the form $\dfrac{C}{x} + \dfrac{D}{x^2} + \dfrac{E}{x - 1}$ where C, D and E are constants to be found.

Answer

Dealing with Improper Fractions 35

In Pure Year 1, we saw that the '**degree**' of a polynomial is the highest power, e.g. a quadratic has degree 2.
An algebraic fraction is **improper** if the degree of the numerator is **at least** the degree of the denominator.

$\frac{x^2-3}{x+2}$ $\frac{x+1}{x-1}$ $\frac{x^3-x^2+3}{x^2-x}$

A partial fraction is still improper if the degree is the same top and bottom.

Questions might take one of two forms:
- Do the division to express as a quotient and a remainder, e.g. $\frac{x+1}{x-1} \quad \rightarrow \quad 1+\frac{2}{x-1}$
- Express as partial fractions, e.g. $\frac{x^2+x}{(x+1)(x-2)} = A + \frac{B}{x+1} + \frac{C}{x-2}$

Reducing to Quotient and Remainder 36

For example, $7 \div 3 = 2$ remainder 1, so $\frac{7}{3} = 2 + \frac{1}{3}$.
In general,
$\frac{F(x)}{\text{divisor}} = Q(x) + \frac{\text{remainder}}{\text{divisor}}$,
where $Q(x)$ is the quotient.

If
$\frac{x^2+5x-9}{x+2} = Ax + B + \frac{C}{x+2}$,
determine A, B, C.

Working

Test Your Understanding 37

Given that

$\frac{3x^4-2x^3-5x^2-4}{x^2-4} \equiv ax^2+bx+c+\frac{dx+e}{x^2-4}, \quad x \neq \pm 2,$

find the constants a, b, c, d, e. (4 marks)

Tip: insert $+0x$ for missing powers before dividing.

Working

Dealing with Improper Fractions 38

Split $\frac{3x^2-3x-2}{(x-1)(x-2)}$ into partial fractions.

Method 1: Algebraic Division

Method 1

Method 2: Using One Identity

Method 2

Express $\dfrac{9x^2+20x-10}{(x+2)(3x-1)}$ in partial fractions.

(4)

Answer

E **6** Given that $\dfrac{2x^2-1}{x^2+2x+1} \equiv A + \dfrac{B}{x+1} + \dfrac{C}{(x+1)^2}$, find the values of the constants A, B and C. **(4 marks)**

Answer

(P) 7 By factorising the denominator, express the following as partial fractions:

a $\dfrac{4x^2 + 17x - 11}{x^2 + 3x - 4}$

Answer

(E) 8 Given that $\dfrac{6x^3 - 7x^2 + 3}{3x^2 + 11x - 10} \equiv Ax + B + \dfrac{C}{3x - 5} + \dfrac{D}{x + 2}$, find the values of the constants A, B, C and D.

Answer

Informal proof of method for repeated factors 43

If a factor is repeated, why do have a partial fraction with the squared and one without?
When we split into partial fractions, we want each fraction to be non-top-heavy algebraic fractions – recall this means that the 'order' of the numerator has to be less than the order of the denominator. We assume the most generic non-top-heavy fraction possible, i.e. where the order of the numerator is one less than the denominator...

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2}$$

Since the denominator is quadratic (order 2), we want the **most generic possible numerator whilst avoiding the fraction being top heavy**, i.e. linear (order 1).
Just putting $\frac{B}{x^2}$ wouldn't be good enough because there *might* have been an x term in the numerator.

order 2

order 1

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{Bx}{x^2} + \frac{C}{x^2}$$

Split the fraction.

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{B}{x} + \frac{C}{x^2}$$

We can see therefore we have both the x with an without the squared in the denominator.

Informal proof of method for repeated factors 44

This is easy enough if the repeated factor is x^2, but what about more general repeated linear factors?

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{Bx+C}{(x+1)^2}$$

We want something in the numerator where a $(x+1)$ will cancel, so a bit of clever manipulation is required.

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B(x+1)-B+C}{(x+1)^2}$$

Split the fraction as before.

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B(x+1)}{(x+1)^2} + \frac{C-B}{(x+1)^2}$$

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{D}{(x+1)^2}$$

$C-B$ is just a generic constant, so replace with a single constant.